

Electrodynamics

Independence Semester 2024

Homework 6

December 1, 2024

Problem 1

a) A hollow right circular cylinder of radius b has its axis coincident with the z -axis and its ends at $z = 0$ and $z = L$. The potential on the end faces is zero, while the potential on the cylindrical surface is given by $V(\varphi, z)$. Using the appropriate separation of variables in cylindrical coordinates, find a series solution for the potential anywhere inside the cylinder.

b) For the cylinder in part a) the cylindrical surface is made of two equal half-cylinders, one at potential V and the other at potential $-V$, so that

$$V(\varphi, z) = \begin{cases} V, & -\frac{\pi}{2} < \varphi < \frac{\pi}{2} \\ -V, & \frac{\pi}{2} < \varphi < \frac{3\pi}{2}. \end{cases} \quad (1)$$

Find the potential inside the cylinder.

Problem 2

A localized distribution of charge has a density $\rho(\vec{r}) = \frac{1}{64\pi} r^2 e^{-r} \sin^2 \theta$.

a) Make a multipole expansion of the potential due to this charge density and determine all the non-vanishing multipole moments. Write down the potential at large distances as a finite expansion in Legendre polynomials.

b) Determine the potential explicitly at any point in space, and show that near the origin, correct to order r^2 ,

$$\phi(\vec{r}) \approx A \left[\frac{1}{4} - \frac{r^2}{120} P_2(\cos \theta) \right]. \quad (2)$$

Find A .

Problem 3

A short piece of wire is placed along the z -axis, centered at the origin. The wire carries a total charge Q , and the linear charge density λ is an even function of z . The r.m.s length of the charge distribution in the wire is ℓ_0 , i.e., $\ell_0^2 = \frac{1}{Q} \int_{\text{wire}} z^2 \lambda(z) dz$.

- a) Find the dipole and quadrupole moments for this charge distribution. Express your p_i and Q_{ij} s in terms of ℓ_0^2 .
- b) Give an expression for the potential $\phi(r, \theta)$ for large r , including all terms through the quadrupole contribution,